

ISTIAQE AHAMED

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Education

M.Sc. in Applied Mathematics and Computational Science May 2025 – December 2026
(Expected)
North South University
CGPA **3.94/4.00** (Till 34/40 credits)

B.Sc. in Industrial & Production Engineering January 2018 – May 2023
Khulna University of Engineering & Technology
CGPA **3.05/4.00** (last 80 credits: **3.27/4.00**)

Research Interests

- Optimization Algorithms and Machine Learning for Large-Scale Systems
- Operations Research
- Decision Making Under Uncertainty
- Supply Chain Resilience
- High Performance Computing

Work Experience

Graduate Research Assistant (GRA) January 2026 – Present
Supervisor: [Dr. Md. Mamun Molla](#)
Department of Mathematics and Physics, North South University

Graduate Teaching Assistant (GTA) September 2025 – Present
Department of Mathematics and Physics, North South University
(Assisting in teaching Linear Algebra, Ordinary Differential Equations, Operations Research; grading assignments and exams)

Research Experience

Master's Thesis (In Progress):

- **High-Performance Computing for Large-Scale Vehicle Routing Problems with Time Windows: Cross-Architecture Parallelization of Swarm Intelligence Metaheuristics**
[GitHub Repository](#) Supervisor: [Dr. Md. Mamun Molla](#)
Key Findings:
 - Introduced the per-particle evaluation cost (τ) as a novel instance-aware predictor of parallel efficiency, resolving the paradox of high parallelizable fractions coexisting with poor speedup for clustered VRPTW instances.
 - GPU-resident CUDA PSO achieved $53\times$ - $657\times$ speedup over the serial CPU baseline and Pareto-dominates all OpenMP configurations on both runtime and monetary cost simultaneously.
 - Demonstrated that the overhead-augmented Amdahl model fails for $\tau < 1\text{ms}$ instances, requiring a new linear throughput model for GPU performance prediction.
 - Proposed a two-component evaluation cost framework ($\tau_{\text{construct}} + \tau_{\text{update}}$) and pheromone synchronisation ratio (γ) for characterizing distributed Ant Colony parallelism.

Journal Papers:

- **Cross-Architecture Scalability and Performance Modeling of OpenMP and CUDA-Based Particle Swarm Optimization for Large-Scale VRPTW**, *Under Review, Swarm & Evolutionary Computation*
[GitHub Repository](#)
Author: Istiaque Ahamed, Dr. Md. Mamun Molla
- **Decision-Calibrated Conformal Buffering for Supplier Delay Risk: A Machine Learning Framework with Drift Adaptation and Mondrian Coverage Equity**, *Under Review, International Journal of Production Research*
[GitHub Repository](#)
Author: Istiaque Ahamed, Dr. Md. Mamun Molla
- **Converging Blockchain, Artificial Intelligence, and Digital Twin Technologies for Flexible Supply Chains: A Dynamic Capability View of Visibility, Predictive Accuracy, and Decision-Making Agility**, *Submitted, Global Journal of Flexible Systems Management*
[GitHub Repository](#)
Author: Istiaque Ahamed, Talat Mahmud Chowdhury

Conference Papers:

- **How Playing Position and Market Era Shape Football Transfer Fees: A Machine Learning and Explainability Study**, *Under Review, Proceedings of 29th International Conference on Computer and Information Technology*
Author: Istiaque Ahamed, MD Samin Yeasar, Altaf Hussain Chowdhury
- **An Investigation of Blockchain Technology and Supply Chain Flexibility as Complements to Supply Chain Integration and Performance: A Dynamic Capability View**
Author: Nihal Rahman Raad, Palash Saha, Istiaque Ahamed, Md. Omar Shadat Sarker, Ruhaniayth Bin Afsar
<https://doi.org/10.46254/BA07.20240025>
- **An Analysis of Barriers to Supply Chain Resilience Adoption in the Bangladesh Food Industries: A Fuzzy TOPSIS Approach** (Secured second place in the Supply Chain and Logistics category)
Author: Ruhaniayth Bin Afsar, Istiaque Ahamed, Md. Omar Shadat Sarker
<https://doi.org/10.46254/BA07.20240045>
- **Modeling the Barriers to Supply Chain Resilience in the Footwear Industry in Bangladesh**
Author: Istiaque Ahamed, Dr. Md. Rafiquezaman, Palash Saha, Ruhaniayth Bin Afsar
[ProQuest Link](#)

Projects

- **Comparative Analysis of Particle Swarm Optimization and Genetic Algorithm for Solving Vehicle Routing Problem with Time Windows and Capacity Constraints** (*Independent Project, January 2026*)
 - Implemented and developed metaheuristic algorithms (PSO and GA) for the Vehicle Routing Problem with time window and capacity constraints, in Python, using real-world size datasets.[GitHub Repository](#)
- **Breast Cancer Diagnosis: Performance Benchmarking and Statistical Validation of Machine Learning Models** (*Academic Project for AMCS 509, December 2025*)
[GitHub Repository](#)
- **Vehicle Routing Optimization using Particle Swarm Optimization** (*Independent Project, April 2025*)
 - Applied PSO to solve a VRP logistics problem, achieving better cost efficiency than a published MILP

solution.

[GitHub Repository](#)

- **Custom Genetic Algorithm-Based Animated Visualization for Multi-Constrained VRP** (*Independent Project, May 2025*)
 - Benchmarked performance against MILP from published literature; while global optimality wasn't reached, the heuristic demonstrated fast, feasible routing for all 24 customers with penalty cost reduction over generations.[GitHub Repository](#)

Awards & Honors

- **First Runner-up, AutoCAD Workshop** - *2019* (month-long competition organized by the IRCC Club of KUET)
- **70% Scholarship Winner, ISCEA PTAK Prize** - *2021* (global supply chain case competition; team secured 70% scholarship for CSCA Certification Course)
- **Second Place, Supply Chain and Logistics Category** - *2024*
7th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management, Dhaka, Bangladesh

Standardized Test Scores

Graduate Record Examination (GRE) - 16 December, 2025

	Verbal	Quant	AWA	Total
Score	152	167	3.5	319
Percentile	48th	76th	43rd	—

Volunteer Experience

- **IEM Robotics & CAD Club**
 - President (2022–2023)
 - Assistant General Secretary (2020–2022)
- **IEOM KUET Student Chapter**
 - Vice President (2022)

Research Skills

- **Metaheuristics:** PSO, GA, ACO
- **Parallel Computing:** OpenMP, MPI, CUDA
- **Programming Languages:** C, Python
- **Libraries:** NumPy, Pandas, Matplotlib

Certifications

- **Python for Data Science, AI & Development** - IBM Corporation (Coursera), May 2025
- **Supervised Machine Learning: Regression and Classification** - DeepLearning.AI & Stanford University (Coursera), September 2025
- **Python** - Kaggle, April 2025
- **Supply Chain Management Specialization** (5 Courses) -Rutgers, The State University of New Jersey (Coursera), June 2020

References

- **Dr. H. M. Belal**
Reader in Operations and Quality Management
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- **Dr. Md. Mamun Molla**
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- **Dr. Md. Rafiquzzaman**
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